

Full length article

EchoType: Identity recognition from typing-induced ultrasonic reflections

Yena So ^{a,1}, Youngjoo Park ^{a,1}, Chaehyeon Kim ^{a,2}, Jihee Lee ^{a,2}, Juhyeon Kim ^{a,2},
Sunwoo Lee ^{b,*}

^a Seoul Women's University, Seoul, Republic of Korea

^b Ewha Womans University, Seoul, Republic of Korea

ARTICLE INFO

Keywords:

Biometrics
Keystroke dynamics
Ultrasound sensing
Identity recognition
Behavioral biometrics

ABSTRACT

Securely recognizing a user's identity without disrupting everyday interactions remains a central challenge in modern computing environments. Prior keystroke-dynamics-based approaches primarily rely on timing patterns extracted from discrete key events and often achieve high accuracy through increasingly complex deep learning models. However, because these methods operate on abstracted event-level features, prior studies have shown that they can be vulnerable to spoofing attacks in which adversaries intentionally mimic keystroke timing patterns. In this paper, we introduce ECHOType, an identity recognition approach that departs from event-driven keystroke modeling and instead leverages ultrasonic reflections generated during ordinary typing. As a user types, continuous finger motion and anatomical structure modulate inaudible ultrasonic signals emitted by a commodity smartphone, producing spatiotemporal reflection patterns that are physically coupled to the user's motion. ECHOType operates in a physical signal space shaped by both behavior and anatomy, rather than relying solely on key-event timing. ECHOType requires no dedicated sensors or modifications to input devices and integrates seamlessly into existing computing environments using only the built-in speaker and microphone of a smartphone. To accommodate diverse usage scenarios, we evaluate ECHOType under both single-user verification and multi-user identification settings. Experimental results across multiple devices, keyboard types, and everyday environments demonstrate robust performance: ECHOType achieves an Equal Error Rate (EER) of 0.031 for single-user verification, a macro F1-score of 0.940 for multi-user identification, and 95.5% accuracy in unknown-user rejection. These results indicate that typing-induced ultrasonic reflections can serve as a practical and deployable identity recognition mechanism without compromising usability in modern computing environments.

1. Introduction

Securely recognizing a user's identity without imposing additional burden on everyday interactions remains a key challenge in modern computing environments. Knowledge-based identity verification mechanisms, such as passwords and PINs, are still widely used; however, when deployed alone, they are vulnerable to credential theft, reuse, and observation attacks. To mitigate these weaknesses, two-factor and multi-factor verification schemes have been introduced. While effective in improving security, many of these approaches require additional verification steps—such as launching separate applications, entering one-time codes, or using external devices—which often degrade usability and hinder seamless integration into real-world systems.

To address these limitations, recent research has explored identity recognition approaches that leverage signals naturally generated from users' everyday behaviors, enabling verification without requiring explicit additional actions beyond the primary login interaction. Among various behaviors, keystroke activity is one of the most fundamental and ubiquitous interactions in modern computing environments and continuously reflects user-specific behavioral characteristics. Prior work on keystroke dynamics has primarily focused on local features such as inter-keystroke timing patterns or pressure-related information (Shadman et al., 2025). However, these approaches face inherent limitations: (1) they often fail to capture the spatial and dynamic complexity of typing behavior, and (2) they frequently rely on specialized

* Corresponding author.

E-mail addresses: yenao0@swu.ac.kr (Y. So), zer0joo@swu.ac.kr (Y. Park), chaehyun04.kim@swu.ac.kr (C. Kim), leejihee1102@swu.ac.kr (J. Lee), jhrx0529@swu.ac.kr (J. Kim), sun.lee@ewha.ac.kr (S. Lee).

¹ Co-first authors.

² These authors contributed equally to this work.

<https://doi.org/10.1016/j.cose.2026.104947>

Received 30 December 2025; Received in revised form 23 March 2026; Accepted 25 April 2026

Available online 8 May 2026

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keyboards or additional sensing hardware, which restricts their deployability in practical settings. Recent studies have sought to overcome these limitations by extending keystroke dynamics with deep learning models that encode timing patterns in higher-dimensional representations (Acien et al., 2021; Stragapede et al., 2024; Budzys et al., 2025; Medvedev et al., 2025). While these approaches have significantly improved recognition accuracy, they remain fundamentally grounded in discrete key events and their associated temporal features. As noted in prior work, such event-driven representations can be vulnerable to spoofing attacks in which adversaries intentionally mimic keystroke timing patterns, even under the common assumption that the sensing channel itself is trusted (Shadman et al., 2025).

In this paper, we introduce ECHOType, an identity recognition approach designed to balance security and usability while remaining easily deployable within existing computing environments. ECHOType identifies users by leveraging ultrasonic reflections generated during ordinary typing, without requiring any modification to users' natural interaction patterns. As a user types, subtle finger motions modulate inaudible ultrasonic signals propagating through the surrounding space, producing reflection patterns that encode both typing behavior and anatomical characteristics of the user's fingers. By operating on these spatially modulated bioacoustic reflections, ECHOType moves beyond purely temporal or event-driven features commonly used in conventional keystroke dynamics, and instead captures continuous physical interactions between the user and the environment. While prior ultrasound-related keystroke studies have predominantly examined ultrasonic signals from an attack-oriented perspective—such as inferring keystroke positions or reconstructing typed content based on signal timing (Zhao et al., 2023)—we take a fundamentally different viewpoint. Rather than treating ultrasonic reflections as an unintended leakage channel, we analyze the biometric information embedded in these signals and repurpose typing-induced ultrasonic reflections as a sensing modality for identity recognition.

Our primary contributions are summarized as follows:

- We propose ECHOType, a novel identity recognition approach that leverages ultrasonic reflection signals naturally produced during everyday typing. By capturing spatiotemporal acoustic characteristics arising from both typing behavior and finger anatomy, ECHOType extends behavioral biometrics beyond conventional timing- and event-driven keystroke features.
- ECHOType does not require dedicated sensors or specialized hardware. Instead, it repurposes a widely available commodity smartphone as a lightweight sensing proxy, enabling seamless integration into existing computing environments. Users interact with the system through their regular typing behavior, without requiring additional actions beyond the primary login interaction.
- We design two identity recognition models tailored to different deployment settings: (1) a single-user verification model that distinguishes a legitimate user from all others, and (2) a multi-user identification model that recognizes multiple authorized users and detects unauthorized access attempts. Together, these models cover security requirements in both personal and shared computing environments.
- We evaluate ECHOType using data collected from 30 participants across keyboard types, and everyday usage environments. Experimental results demonstrate robust performance, achieving an Equal Error Rate (EER) of 0.031 in single-user verification, a macro F1-score of 0.940 in multi-user identification, and 95.5% accuracy in unknown-user rejection. These results indicate that ECHOType can effectively detect unauthorized access attempts and serve as a practical identity recognition mechanism.

2. Related works

We review two major research domains relevant to our work: (1) identity recognition using keystroke dynamics, and (2) sound- or ultrasound-based analysis of typing activities. While both lines of work observe users' typing behavior, they differ fundamentally in sensing mechanisms and in the type of information leveraged for identity recognition.

2.1. Keystroke-dynamics-based identity recognition

Keystroke dynamics is a well-established behavioral biometric that characterizes users based on their typing habits. Early studies primarily relied on temporal features such as dwell time, inter-key latency, and key press-release intervals (Shadman et al., 2025). Representative examples include the anomaly-detection benchmark by Killourhy and Maxion (2009), Gaussian latency modeling by Giot et al. (2009, 2012), and vector-difference-based approaches by Bakelman et al. (2013). These methods demonstrate that reasonable verification performance can be achieved using simple statistical models, but they inherently represent typing behavior using low-dimensional temporal summaries. As a result, fine-grained biomechanical and anatomical characteristics underlying finger motion are largely overlooked. To enrich the feature space, one line of work introduced pressure sensors or instrumented keyboards to capture tactile information such as key pressure and contact area (Grabham and White, 2008). However, such approaches require specialized hardware and are difficult to deploy across diverse real-world environments.

Recent advances in deep learning have further improved keystroke-dynamics-based user recognition by modeling high-dimensional temporal dependencies. For example, Acien et al. employed LSTM networks (Acien et al., 2021), Stragapede et al. proposed a transformer-based architecture (TypeFormer) (Stragapede et al., 2024), and Budzys et al. combined interpolation-based data fusion with Siamese neural networks (Budzys et al., 2025). Medvedev et al. reported extremely low EERs using Siamese-network-based frameworks (Medvedev et al., 2025). While these methods significantly advance identity recognition performance, they still rely exclusively on timing patterns derived from discrete key events and require complex model architectures and data encodings. Meanwhile, Kim et al. proposed a model that transforms keystroke patterns into graph-based image representations for user recognition (Kim et al., 2025). This work is notable in that it emphasizes the interpretability of data representation rather than relying on increasingly complex model architectures. However, the method is designed for mobile touch-based input environments and leverages features such as pressure, finger contact area, and touch location, which limits its applicability to physical keyboard settings.

Prior work has also noted the vulnerability of keystroke-dynamics-based systems to spoofing attacks, particularly those that mimic timing patterns (Shadman et al., 2025). This vulnerability arises because the underlying features are event-driven and temporally abstracted. In contrast, ECHOType captures ultrasonic reflection patterns generated by continuous finger motion, enabling the extraction of both behavioral and anatomical characteristics beyond event-driven timing features.

2.2. Sound- and ultrasound-based keystroke analysis

Another line of research analyzes acoustic or ultrasonic signals generated during typing, primarily from an adversarial perspective. Early works extracted MFCCs or mel-spectrograms from typing sounds to classify or reconstruct keystrokes (Halevi and Saxena, 2012; Wang et al., 2016; Anand and Saxena, 2018; Compagno et al., 2017). Subsequent studies leveraged deep audio models (Giallanza et al., 2019) or layout-agnostic localization techniques, as demonstrated by KeystrokeSniffer (Huang et al., 2024). More recently, ultrasound-based

attacks such as UltraSnoop (Zhao et al., 2023) exploited time-of-arrival differences between microphones to localize keystrokes in the ultrasonic band. These approaches share two common characteristics: (1) they treat typing-induced acoustic signals as a leakage channel for inferring typed content or key locations, and (2) they rely heavily on geometric or timing relationships between microphones. ECHOType differs from these prior works in how ultrasonic signals are interpreted and utilized. Rather than inferring keystrokes or recovering input content, we repurpose ultrasonic reflections as a biometric signal for identity recognition. Unlike prior approaches such as UltraSnoop (Zhao et al., 2023), which rely on geometric or timing-based features (e.g., time-of-arrival differences) to localize keystrokes, our method analyzes the full reflection patterns induced by continuous finger motion. Specifically, we transform ultrasonic signals into spectrogram-based representations that capture fine-grained spatiotemporal variations, enabling the extraction of both behavioral and anatomical characteristics. Operating in the inaudible ultrasonic band (above 20 kHz), ECHOType can be implemented using commodity smartphone hardware while introducing a new sensing modality for identity recognition.

3. Motivation

Typing generates subtle physical motions that vary across individuals due to differences in finger kinematics, hand posture, and typing rhythm. These fine-grained characteristics reflect fine-grained behavioral patterns shaped by individual finger motion and posture of the user, yet they are difficult to capture using conventional timing-based keystroke features alone (Shadman et al., 2025). This raises a natural question: *can alternative physical signals generated during everyday typing encode richer user-specific information suitable for identity recognition?*

To explore this question, we investigate how finger motion modulates high-frequency acoustic signals during typing. Fig. 1-(a) illustrates ultrasonic reflections recorded by a commercial smartphone while a participant typed on a standard keyboard. In our setup, the smartphone continuously emitted inaudible 21.5 kHz ultrasound from its top speaker, while both built-in microphones recorded the reflected signals at a 48 kHz sampling rate. As the user typed, subtle finger movements introduced small variations in the amplitude and phase of the returning signals, producing reflection patterns influenced by the user's motion dynamics and hand structure. A 15 kHz high-pass filter was applied to isolate ultrasonic components from ambient audio signals. To further demonstrate the user-specific nature of these reflections, two participants were asked to type the same phrase, "hello world", using their natural typing styles. As shown in Fig. 1-(b), Participant A's multi-finger touch-typing resulted in rapidly fluctuating signal amplitudes, whereas Participant B's hunt-and-peck typing produced smoother temporal variations. Notably, even when pressing identical keys, the reflected waveforms differed across participants. This observation suggests that anatomical factors – such as finger length, joint movement, and hand alignment – also contribute to the modulation of ultrasonic signals. The corresponding spectrograms in Fig. 1-(c) further illustrate clear temporal and spectral distinctions between users.

These observations motivate the design of ECHOType, which leverages typing-induced ultrasonic reflections captured by a commodity smartphone as a lightweight sensing proxy. By exploiting ultrasonic signals naturally generated during everyday typing, ECHOType enables unobtrusive identity recognition without requiring dedicated sensing hardware or altering existing user workflows.

4. Threat model

We consider an adversary whose goal is to impersonate a legitimate user during identity recognition by exploiting knowledge of the user's credentials, observable typing behavior, or previously captured acoustic signals. We focus on identity recognition scenarios where the system

verifies or identifies a user based on behavioral and biometric signals during normal typing interactions.

Adversary Capabilities. We consider the following realistic attack capabilities, which align with common threat scenarios in practical identity recognition systems:

- *Credential Knowledge:* The attacker knows the victim's correct password but does not possess the victim's biometric characteristics or intrinsic typing-induced acoustic patterns. This models scenarios such as password leakage, reuse, or phishing.
- *Typing-Behavior Observation and Mimicry:* The attacker can observe the victim's typing behavior – either directly or via video recordings – and attempts to imitate the victim's typing dynamics. This includes reproducing finger motion patterns, typing rhythm, hand posture, and overall input cadence. To reflect realistic imitation attempts, we assume the attacker may leverage individuals with similar physical characteristics (e.g., hand size or finger length) to better approximate the victim's typing behavior.
- *Replay Capability:* The attacker has access to previously recorded ultrasonic reflections generated during a legitimate identity recognition session and attempts to replay these signals using external audio playback devices in order to spoof the system.

Adversary Limitations. The attacker cannot modify, compromise, or gain privileged access to the smartphone hardware, ultrasound emission mechanism, or internal signal processing pipeline. The attacker also has no control over the sensing device placement or the surrounding acoustic environment during identity recognition.

5. ECHOType system framework

5.1. Overview

We propose ECHOType, an ultrasound-based identity recognition system designed for seamless integration into existing computing environments. ECHOType exploits the interaction between inaudible ultrasonic signals emitted by a commodity smartphone and the user's finger movements during typing. As the user types, finger motion and individual finger anatomy modulate the reflected ultrasonic signals, producing distinctive spatiotemporal patterns associated with the user. These reflections capture user-specific typing dynamics.

Fig. 2 illustrates an example deployment of EchoType. The system consists of four main components. (1) User Enrollment (Section 5.2), where ultrasonic reflections generated during typing are collected to establish user-specific reference patterns; (2) Ultrasound Emission and Capture (Section 5.3), which emits inaudible ultrasonic signals using the smartphone's built-in speaker and records the reflected signals during typing; (3) Signal Processing and Feature Extraction (Section 5.4), which analyzes the reflected signals to extract discriminative features that jointly encode typing behavior and hand-related structural characteristics; and (4) Identity Recognition (Section 5.5), which interprets the extracted features to perform user verification and identification under both single-user and multi-user scenarios.

5.2. User enrollment

When setting up the identity recognition system for the first time, users undergo an enrollment process to register their typing-induced acoustic signatures. During enrollment, as the user types naturally, a commodity device (e.g., a smartphone) captures the reflected ultrasonic signals in real time. These signals are then processed to construct a user-specific reference model, which serves as a baseline for subsequent identity recognition.

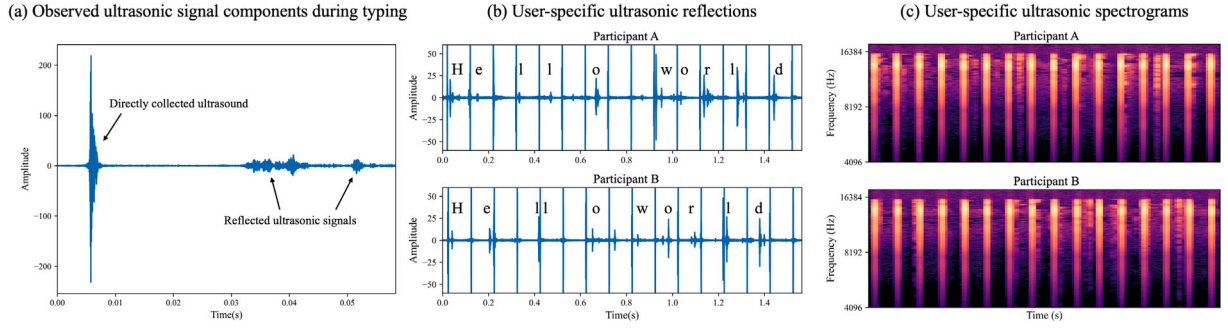


Fig. 1. Ultrasonic reflections generated during typing.



Fig. 2. Physical configuration of ECHOType.

5.3. Ultrasound emission and capture

A standard smartphone serves as both emitter and receiver. Let $s(t)$ denote a sinusoidal ultrasonic carrier centered at $f_c = 21.5$ kHz, emitted from the top speaker. When a user types, each finger motion $m_i(t)$ modulates the reflected wave according to its displacement and velocity. The reflection arriving at the i th path can be expressed as:

$$r_i(t) = A_i(t) s(t - \tau_i(t)) = A_i(t) \cos\left(2\pi f_c t - 2\pi f_c \frac{2m_i(t)}{c}\right),$$

where $A_i(t)$ is the time-varying reflection amplitude, $\tau_i(t) = 2m_i(t)/c$ is the round-trip delay, and c is the speed of sound. The received composite signal is

$$r(t) = \sum_i r_i(t) + n(t),$$

where $n(t)$ captures ambient noise and residual audible keystroke sounds.

Reflections are sampled at 48 kHz using both top and bottom microphones, ensuring full coverage of the ultrasonic band up to 24 kHz. Continuous pulse emission at 0.1-s intervals captures quasi-stationary movement patterns across consecutive keystrokes, providing temporal continuity for behavioral modeling. Because all operations occur beyond the human audible threshold, the process is imperceptible and does not interfere with normal typing.

5.4. Signal processing and feature extraction

Raw ultrasonic reflections include both user-specific modulation and extraneous environmental noise. To isolate the relevant information, we apply a high-pass filter $H(f)$ with cutoff frequency 15 kHz:

$$x(t) = F^{-1}\{H(f)F\{r(t)\}\}.$$

This step eliminates low-frequency audible components, such as mechanical key clicks and speech, while retaining only modulated ultrasound.

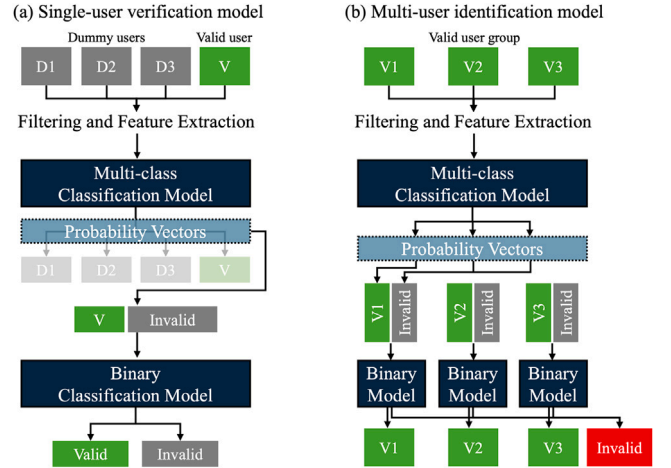


Fig. 3. Architecture of ECHOType for single-user and multi-user model.

Next, each filtered signal $x(t)$ is transformed via the Short-Time Fourier Transform (STFT):

$$X(t, \omega) = \sum_{n=0}^{N_{\text{fft}}-1} x[n + th_l] w[n] e^{-j\omega n},$$

where $w[n]$ is a Hamming window, $N_{\text{fft}} = 2048$, and the hop length $h_l = 512$. The corresponding power spectrum

$$S(t, \omega) = |X(t, \omega)|^2$$

represents the energy distribution across time and frequency, effectively capturing transient finger motions and resonance shifts induced by joint articulation and hand geometry. Each spectrogram is normalized, resized to $(224 \times 224 \times 3)$, and treated as an RGB image $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$ for feature learning.

5.5. Identity recognition

Let $\phi_\theta(\mathbf{I})$ denote the convolutional encoder parameterized by θ , producing a latent representation $\mathbf{z} = \phi_\theta(\mathbf{I}) \in \mathbb{R}^d$. The encoder extracts local-temporal correlations corresponding to micro-motion patterns and spatial energy flows.

5.5.1. Single-user model

In a single-user deployment, identity recognition is formulated as a single-user verification problem, where the system must accept the enrolled user while rejecting all others. Although the final decision is binary, auxiliary data from non-target users (i.e., dummy users) are incorporated during representation learning to improve feature separation and robustness to imitation. The encoder is first trained as a multi-class CNN using spectrograms from both the target user and

auxiliary dummy users. Given a sample $\mathbf{I}$, the CNN outputs a probability vector $\mathbf{p} = \text{softmax}(W_c \mathbf{z} + b_c) \in \mathbb{R}^K$, which reflects the distribution of latent acoustic features over K pseudo-classes. This intermediate representation is then evaluated by a binary Support Vector Machine (SVM):

$$g(\mathbf{p}) = \text{sign}(\mathbf{w}^\top \mathbf{p} + b),$$

yielding an accept/reject decision.

This hybrid CNN–SVM pipeline expands the discriminative space of the one-class model, enabling effective detection of unseen impostors and adaptive imitators. The overall architecture of the single-user model is illustrated in Fig. 3-(a). The CNN consists of two convolutional blocks, each composed of two Conv2D layers with 32 and 64 filters, respectively, using 3×3 kernels and ReLU activations. Each block is followed by max-pooling, batch normalization, and dropout (rate 0.2). The resulting feature maps are flattened and passed through a fully connected layer with 128 neurons (ReLU activation), followed by a softmax layer. The softmax output is then used as the input feature vector for the SVM classifier, which performs the final decision.

This two-stage design decouples representation learning from decision boundary construction, allowing the model to generalize beyond seen users while maintaining strict rejection behavior for unknown inputs. The detailed pseudocode of the single-user model is provided in Appendix (see Algorithm 1).

5.5.2. Multi-user model

In multi-user environments, the task becomes a 1: N open-set recognition problem. A shared encoder ϕ_θ is retained, but the final classification layer is replaced by a set of per-user binary SVMs $\{B_i\}_{i=1}^N$. The softmax output $\mathbf{p}$ of the CNN encoder is used as the input feature vector for each user-specific SVM classifier. Each classifier produces a decision score $g_i(\mathbf{p})$ indicating the likelihood that the input belongs to user i . The decision rule is:

$$\hat{i} = \begin{cases} \arg \max_i g_i(\mathbf{p}), & \text{if } \max_i g_i(\mathbf{p}) > t_i, \\ \text{unknown}, & \text{otherwise.} \end{cases}$$

This open-world thresholding prevents the model from forcibly assigning an impostor to the nearest known user, improving robustness to unseen individuals. The overall architecture of the multi-user model is illustrated in Fig. 3-(b). The detailed pseudocode of the multi-user model is provided in Appendix (see Algorithm 2).

5.5.3. Model optimization and interpretation

For all SVMs, the hyperparameters (C , kernel type, and γ) are optimized through cross-validation on balanced data subsets. Radial Basis Function (RBF) kernels achieve the best trade-off between margin separation and generalization performance. During training, each user's spectrogram samples are augmented with additive Gaussian noise to simulate minor acoustic perturbations, thereby mitigating the impact of replay artifacts and minor environmental variations. Feature saliency analysis reveals that the CNN encoder predominantly attends to localized spectral regions corresponding to finger trajectories and joint reflections, indicating that both anatomical and behavioral cues are implicitly encoded. This observation suggests that the learned representation $\mathbf{z}$ effectively captures intrinsic, user-specific dynamics that are difficult to precisely replicate through imitation or simple replay under realistic deployment constraints.

5.5.4. Design rationale

The hybrid CNN–SVM framework is motivated by the open-set nature of ECHOTYPE, which requires both user identification and rejection of unknown users. The CNN learns user-specific representations from spectrogram inputs, and SVMs form decision boundaries on top of these representations. Unlike metric learning approaches that rely on distance-based similarity, this design provides an explicit mechanism for user-wise decision making, which aligns naturally with open-set recognition.

6. Experimental setup

To evaluate the feasibility and robustness of the proposed ultrasonic keystroke-based identity recognition framework, ECHOTYPE, we conducted controlled experiments using a commodity smartphone and standard typing interfaces. The experimental setup was designed to capture user-specific spatiotemporal echo patterns during natural typing behavior.

Device Configuration. Experiments were performed using an iPhone 13 Pro Max, which served as both an ultrasonic emitter and receiver. The device is equipped with two microphones (top and bottom) and stereo speakers capable of reproducing ultrasound up to 22 kHz. The phone was positioned horizontally at the center of a standard membrane keyboard – chosen for its ubiquity in everyday use – with its lower edge 5 cm above the keyboard surface and tilted by approximately 54° so that the screen faced downward toward the typing area. This configuration was selected based on preliminary experiments to provide sufficient acoustic coverage of both hands and stable reflection paths for ultrasonic sensing. We fixed this configuration across all experiments to isolate user-specific signal variations from geometric effects, rather than optimizing placement per user.

Software Implementation. A custom mobile application was developed using Apple's Xcode environment. The app emits continuous-wave ultrasound centered at 21.5 kHz from the top front speaker while recording incoming reflections via both microphones at a 48 kHz sampling rate. According to the Nyquist–Shannon sampling theorem (Nyquist, 1928), this sampling rate is sufficient for lossless acquisition of the ultrasonic band. Raw waveforms were stored in 16-bit PCM format and processed offline using Python for filtering, time–frequency transformation, and model construction.

Participants and Data Collection. We recruited 30 participants (aged 20–40, 70% female, 30% male), an age range representing typical computer users (Statista Research Department, 2025). Each participant was fully briefed on the experiment's purpose and provided written informed consent in accordance with institutional ethical requirements. Participants received a coffee voucher as compensation. During data collection, each participant was instructed to type randomly assigned passwords (e.g., “Tjgklvsb!”) and short natural phrases (e.g., “hello world”) while maintaining their normal typing posture and rhythm. Ultrasonic reflections were recorded concurrently with typing, yielding 120 valid samples per participant during the initial session. Invalid entries – such as mistyped passwords – were discarded. Collected samples were randomly divided into training and testing sets with an 80:20 split. For each participant, training and testing samples were strictly separated, ensuring that no sample overlap exists between the two sets and preventing data leakage. This setup follows standard evaluation protocols in biometric recognition systems. To further evaluate open-set recognition, unseen users were introduced only during evaluation and were not included in the training process. The first session required approximately one hour per participant (including rest periods), while the follow-up session required around 15 min. To facilitate later behavioral correlation analysis, each session was video recorded with participant consent, ensuring synchronization between keystroke actions and corresponding echo signals.

Model Construction and Parameter Tuning. Two sets of identity recognition models were constructed: single-user and multi-user. In the single-user scenario, dummy users are randomly chosen among participants, and the remaining were treated as valid users. Dummy users were selected solely to stabilize representation learning and were not included in evaluation. For CNN training, we used a batch size of 32, 10 training steps per epoch, and 20 epochs in total. Hyperparameter optimization for the SVM classifier was performed using a grid search over the parameter space: kernel type $\in \{\text{linear}, \text{rbf}, \text{poly}\}$, regularization $C \in \{0.1, 1, 10, 100\}$, and kernel coefficient $\gamma \in \{1, 0.1, 0.01, 0.001\}$. Example optimal configurations included `kernel=poly`, $C = 0.1$, and $\gamma = 1$ for a specific single-user model. For the multi-user scenario, we

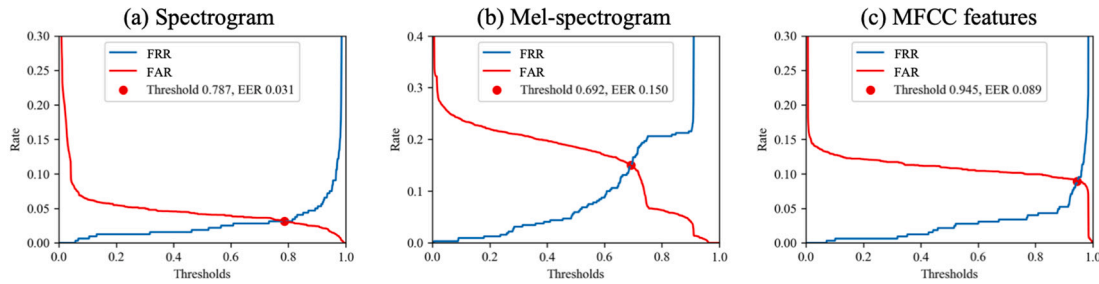


Fig. 4. Comparison of identity recognition performance using different feature representations in the single-user scenario.

formed groups of three valid users to evaluate group-level classification and unknown-user rejection. The CNN encoder structure and SVM tuning process were identical to those used in the single-user model, with per-user SVM classifiers trained for open-world decision boundaries. All models were evaluated using 5-fold cross-validation to ensure statistical consistency.

Ethical and Privacy Considerations. This study was approved by the Institutional Review Board (IRB) of our institution. All participants were anonymized, and their raw ultrasonic data were used solely for model training. Once the models were generated, all raw signals and intermediate feature representations were permanently deleted to prevent biometric data reconstruction. Dummy user data used to stabilize model training were similarly anonymized and irreversibly separated from participant identifiers.

7. Performance metrics

To quantitatively evaluate identity recognition performance, we defined metrics suitable for both closed-set (single-user) and open-set (multi-user) conditions. The selected metrics jointly capture usability (low false rejection), security (low false acceptance), and open-world robustness (accurate rejection of unseen users).

Single-user Metrics. For the single-user scenario, we evaluate performance using the Equal Error Rate (EER). EER corresponds to the operating point at which the false rejection rate (FRR) equals the false acceptance rate (FAR), and is widely used to assess the trade-off between usability and security in verification systems. Lower EER values indicate stronger discriminative capability and more reliable user identity recognition.

Multi-user Metrics. In the multi-user open-world setting, we evaluate two primary aspects: (1) classification accuracy for valid (known) users, measured by the F1-score, and (2) rejection performance for unseen (unknown) users, measured by detection accuracy.

- **F1-score:** We use the F1-score to measure classification performance for enrolled (known) users by balancing precision and recall. It is defined as

$$\text{F1-score} = 2 \times \frac{P \times R}{P + R}, \quad \text{where } P = \frac{TP}{TP + FP} \text{ and } R = \frac{TP}{TP + FN}.$$

- **Detection Accuracy:** We measure unknown-user rejection performance using detection accuracy, defined as the proportion of correctly rejected unknown inputs:

$$\text{Detection Accuracy} = \frac{TR}{TR + FR},$$

where TR and FR denote true and false rejections, respectively.

In summary, the F1-score measures recognition consistency across enrolled users, while detection accuracy complements it by capturing the system's ability to correctly reject inputs from previously unseen individuals. We adopt detection accuracy as a primary open-set metric because it directly reflects practical identification performance under fixed operational thresholds, which aligns with real-world deployment requirements.

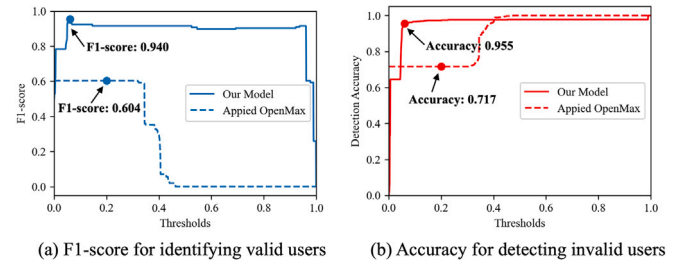


Fig. 5. Comparison of identity recognition performance in open-world multi-user scenarios.

8. Evaluation

8.1. Comparative verification of the model

8.1.1. Effectiveness of spectrogram-based features

We first evaluated the effectiveness of the proposed spectrogram-based feature representation by comparing it with two conventional acoustic features: mel-spectrogram and Mel-Frequency Cepstral Coefficients (MFCC). Each participant typed the arbitrary password “Tjgk-lvsl!” while the corresponding ultrasonic reflections were recorded. After filtering, we extracted three types of features and trained models accordingly. Spectrogram and mel-spectrogram, being two-dimensional representations, were processed using a CNN-2D model, whereas MFCC – being a one-dimensional temporal feature sequence – was processed using a CNN-1D model to preserve its original temporal structure. To account for temporal variability in the MFCC features, we additionally applied Dynamic Time Warping (DTW) (Muda et al., 2010).

Fig. 4 presents the Equal Error Rates (EERs) achieved by each feature set. The mel-spectrogram and MFCC+DTW models yielded EERs of 0.150 and 0.089, respectively, while our spectrogram-based model achieved a substantially lower EER of 0.031. This demonstrates that traditional features optimized for human speech – typically confined to low-frequency ranges – are less effective for ultrasonic analysis. In contrast, the proposed spectrogram representation captures fine-grained temporal and spatial variations in the ultrasonic reflections, resulting in more accurate and stable identity verification.

8.1.2. Open-world detection performance

We next evaluated the robustness of our system in a multi-user open-world scenario, comparing it with a CNN-2D model integrated with OpenMax—an open-set recognition method designed for image classification (Bendale and Boulton, 2016). Both models were trained on spectrogram images derived from filtered ultrasonic reflections. The training set contained samples from three valid users, while the test set included 27 additional participants as unseen impostors.

As shown in Fig. 5(a) and (b), decision thresholds were selected on validation data to balance acceptance of enrolled users and rejection of unseen impostors. Under this setting, the OpenMax-based model

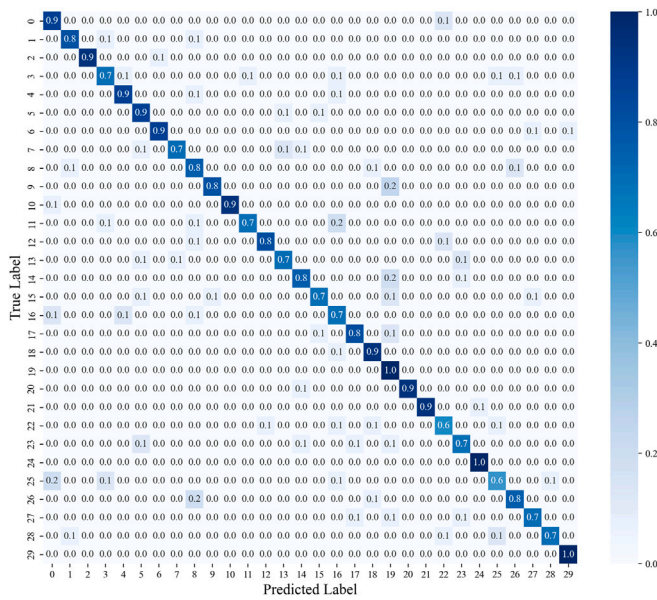


Fig. 6. User distinguishability from ultrasonic typing patterns.

achieved a macro F1-score of 0.604 and a detection accuracy of 0.717 at a threshold of 0.200, indicating limited robustness to unseen users. In contrast, our model achieved a macro F1-score of 0.940 and a detection accuracy of 0.955 at a threshold of 0.060, demonstrating substantially improved recognition of valid users while effectively rejecting impostors. These results highlight the effectiveness of our hybrid CNN-SVM architecture for open-world identity recognition.

8.2. Distinguishability of typing patterns

To assess whether users exhibit distinguishable typing characteristics even when entering the same password, we conducted a multi-class classification experiment. Thirty participants were asked to type the identical password while ultrasonic reflections were captured using a commercial smartphone. After filtering, spectrograms were extracted and fed into a CNN-2D classifier. The dataset was divided into a training set of shape (1920, 224, 224, 3) and a test set of shape (480, 224, 224, 3). Fig. 6 shows the confusion matrix visualized as a heatmap. The model achieved an overall classification accuracy of 81.40%, with strong diagonal dominance indicating clear separability among users. This confirms that typing-induced ultrasonic reflections encode user-specific spatiotemporal characteristics, enabling biometric distinction even under identical input content. Such discriminability supports the feasibility of integrating ultrasound-based keystroke dynamics as a practical biometric modality.

8.3. Overall identity recognition performance

We evaluate the overall performance of the proposed ECHOType framework under both single-user and multi-user conditions, examining robustness to environmental, temporal, and physical variations.

8.3.1. Single-user scenario

Three dummy users and one valid user participated in each single-user experiment. All users typed the same password under ambient office noise (approximately 60 dB), which included background keyboard sounds and low-level conversation. The valid user's typing data were first collected without any accessories on the hands or wrists, followed by two additional sessions designed to assess robustness: (1) while wearing everyday accessories such as a wristwatch, bracelet, or

ring, and (2) two weeks after the initial session to evaluate temporal stability. Fig. 7-(a) summarizes the results across these conditions. When no accessories were worn, the model achieved an Equal Error Rate (EER) of 0.031 at a decision threshold of 0.787, confirming high discrimination performance under typical conditions. When accessories were introduced, the FRR increased to 0.076, suggesting that metallic or reflective surfaces slightly alter ultrasonic propagation and reflection paths. After two weeks, the FRR rose modestly to 0.061, indicating minor temporal drift due to behavioral adaptation or environmental change. Despite this degradation, ECHOType maintained substantially higher stability and accuracy than conventional timing-based keystroke methods.

8.3.2. Multi-user scenario

For the multi-user evaluation, each group consisted of three valid users. As in the single-user setup, participants typed identical passwords and natural text while maintaining their usual typing behavior under three conditions: (1) ambient noise (60 dB), (2) with accessories, and (3) two weeks after the initial session. Fig. 7-(b) summarizes the multi-user identification performance of ECHOType under diverse real-world variations. The model achieved a macro F1-score of 0.940 in the presence of ambient noise, 0.861 when hand accessories were worn, and 0.906 for data collected two weeks later. These results indicate that the system maintains stable discriminative capability across environmental and temporal changes, as well as under minor physical variations. Such robustness demonstrates its feasibility for multi-user identity recognition in everyday computing environments.

8.3.3. Keyboard robustness

We further evaluated the system's robustness across different keyboard mechanisms under ambient office noise conditions. Compared with the membrane keyboard baseline (EER = 0.031 for single-user login and macro F1 = 0.940 for multi-user identification), both scissor-switch and mechanical keyboards resulted in modest performance degradation. Specifically, under the same decision threshold, the single-user FRR increased to 0.081 for the scissor-switch keyboard and 0.089 for the mechanical keyboard. In the multi-user scenario, the macro F1-score decreased to 0.862 and 0.828 for the scissor-switch and mechanical keyboards, respectively. This observed degradation can be attributed to subtle biomechanical variations in finger motion – such as differences in key travel distance, surface height, and typing posture – introduced by the structural characteristics of each keyboard type, rather than changes in the acoustic environment itself. When fine-tuned using a small amount of keyboard-specific data (approximately 10% of newly collected samples), performance recovered to an FRR of 0.043 for the scissor-switch keyboard and 0.047 for the mechanical keyboard in the single-user scenario, and to macro F1-scores of 0.912 and 0.898, respectively, in the multi-user scenario. This demonstrates lightweight adaptability across heterogeneous keyboard interfaces.

8.4. Impact of model parameters on performance

8.4.1. Number of users

We evaluated how identity recognition performance varies with the number of users in both single-user and multi-user configurations. All decision thresholds were selected based on a separate validation dataset. In the single-user scenario, the Equal Error Rate (EER) was 0.029 with four dummy users (threshold = 0.691), 0.030 with five dummy users (threshold = 0.128), and 0.062 with six dummy users (threshold = 0.832). Although the lowest EER was observed with four dummy users, the improvement over configurations with fewer dummy users (e.g., three) was marginal. Increasing the number of dummy users inevitably raises model complexity and training cost. Therefore, we selected three dummy users as a balanced configuration that maintains lightweight operation without degrading verification accuracy. This trade-off suggests that beyond a certain point, adding auxiliary users

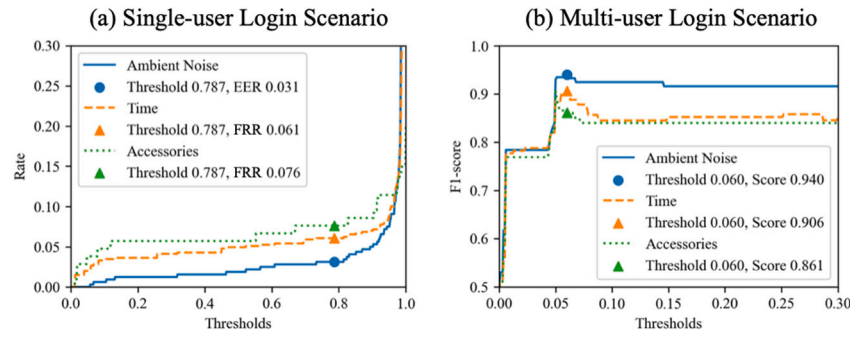


Fig. 7. User identity recognition performance under different conditions across single-user and multi-user scenarios.

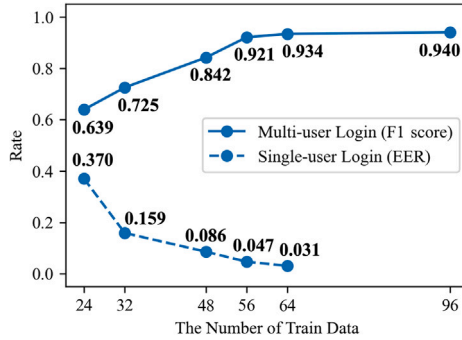


Fig. 8. User identity recognition performance based on the size of the training dataset.

yields diminishing returns for single-user verification. In the multi-user scenario, the macro F1-scores were 0.908, 0.897, and 0.871 for four, five, and six valid users, respectively (thresholds = 0.035, 0.029, and 0.049). Performance exhibited a gradual decline as the number of enrolled users increased. This trend reflects a well-known characteristic of multi-user scenario, where identification becomes increasingly challenging as the number of classes grows. From a deployment perspective, organizing users into smaller, modular groups can help maintain stable performance while improving scalability and interpretability of the identity recognition system.

8.4.2. Training dataset size

We further analyzed the impact of training dataset size on identity recognition performance, examining the trade-off between usability (shorter enrollment duration) and robustness. For both single-user and multi-user settings, models were trained using 24, 32, 48, 56, and 64 samples per user. As shown in Fig. 8, performance consistently improved as the number of training samples increased and began to stabilize beyond approximately 56 samples. When fewer than 48 samples were available, performance degradation was observed, reflecting insufficient coverage of natural intra-user variability—a common challenge in behavioral biometric systems. To assess practical enrollment overhead, we consider an average typing speed of approximately 52 words per minute (WordsRated, 2024). Under this estimate, collecting 56–64 password entries would require roughly one minute per user, indicating that the enrollment process remains lightweight while providing stable performance.

8.5. Attack robustness evaluation

We evaluate the robustness of ECHOType against three representative adversarial scenarios derived from our threat model: credential-based attacks, keystroke-mimicking attacks, and replay attacks. Each scenario reflects a distinct attacker capability, ranging from knowledge of the

correct password to attempts at behavioral imitation or signal reuse. Our evaluation focuses on two core aspects of biometric robustness: (i) the system's ability to generalize to legitimate but previously unseen input variations, and (ii) its resistance to adversarial attempts that seek to imitate or reuse typing-induced acoustic signals.

8.5.1. Credential-based attack

In a credential-based attack, the adversary possesses the correct password but does not attempt to replicate the legitimate user's typing behavior. This setting corresponds to common real-world scenarios involving password leakage or credential reuse. Under this condition, ECHOType maintains clear discrimination between genuine users and attackers. As shown in Fig. 9(a), the single-user configuration achieved a FAR of 0.025 at the operational threshold of 0.787, consistent with the baseline results reported in Fig. 7(a). In the multi-user configuration (Fig. 9(b)), the system achieved an attack detection accuracy of 0.943 at a threshold of 0.060, matching the performance observed in Fig. 7(b). These results suggest that knowledge of the correct credential alone is insufficient to compromise the proposed identity recognition system, as ECHOType bases its decisions on typing-induced ultrasonic signatures rather than textual input correctness.

8.5.2. Typing-behavior mimicry attack

In a typing-behavior mimicry attack, the adversary attempts to replicate the legitimate user's typing behavior after observing their typing actions and physical characteristics, such as hand size. Specifically, the attacker aims to bypass the identity recognition system by reproducing finger movements, typing speed, and overall input rhythm using a collaborator with similar physical attributes. To model a realistic mimicry scenario, we selected participants with hand structures comparable to those of the target users and provided them with videos of the targets' typing sessions. These participants were instructed to imitate the observed motions and typing cadence as closely as possible. As shown in Fig. 9(a), the single-user model exhibited a FAR of 0.067 under such mimicry attacks. Although this value is higher than that observed in credential-only attacks, it remains well below the operational decision threshold. In the multi-user scenario (Fig. 9(b)), the system maintained a high attack detection accuracy of 0.903. These results suggest that even when attackers exhibit similar typing styles and physical characteristics, precisely reproducing the ultrasonic reflection patterns captured by ECHOType remains challenging. ECHOType implicitly leverages subtle spectrotemporal cues arising from joint articulation, fine-grained finger micro-motions, and spatial acoustic interactions shaped by individual hand geometry. Such cues are difficult to replicate through observation or deliberate imitation alone.

8.5.3. Replay attack

In a replay attack, the adversary attempts to spoof the system by injecting prerecorded ultrasonic reflections obtained from a legitimate identity recognition session. Because ECHOType emits and receives ultrasound in real time, the reflected waveforms must remain physically

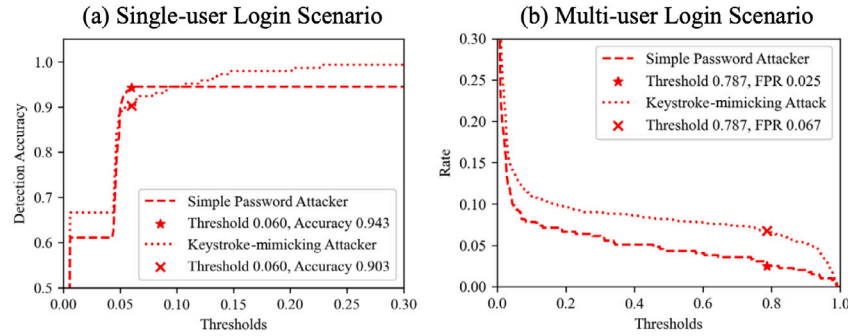


Fig. 9. Performance under different attack scenarios: false acceptance rates (FARs) in the single-user scenario and detection accuracy in the multi-user scenario.

consistent with the user's instantaneous finger motion during typing. Replayed signals cannot reproduce the correct phase evolution and dynamic modulation patterns associated with live interactions. In our experiments, the single-user model exhibited a FAR of 0.0 at the standard operating threshold of 0.787, while the multi-user model achieved an attack detection accuracy of 1.0 at a threshold of 0.060. These results indicate that replayed ultrasonic signals are ineffective under the evaluated threat model. The tight real-time coupling between ultrasound emission and user motion makes it difficult for prerecorded signals to align with the live acoustic feedback loop required for successful identity recognition.

9. Discussion

Device Compatibility and Placement Sensitivity. Our framework was implemented using an iPhone 13 Pro Max, which emitted 21.5 kHz ultrasound and recorded reflections at a 48 kHz sampling rate. Accordingly, any device equipped with a speaker-microphone pair capable of operating in the ultrasonic band and sampling at or above 48 kHz could potentially support ECHOType. This includes many modern Android smartphones, tablets, and even laptops, making the approach broadly deployable without hardware modification. Our observations suggest that the system can tolerate small variations in device placement, such as position changes on the order of centimeters (e.g., around 1 cm) and modest orientation changes (e.g., within approximately 10°). Larger changes in position or orientation, however, may alter acoustic reflection patterns and lead to shifts in the input signal distribution. In such cases, re-collection of reference data and re-training may be necessary to maintain reliable performance.

Use of Dummy Users. In the single-user model, three dummy users were introduced during training to enhance boundary discrimination in the multi-class representation learning stage. While this strategy improves robustness against unseen inputs, collecting real data from dummy participants may be impractical in deployment scenarios due to privacy concerns or limited data availability. As a potential future direction, representative dummy-user data could be synthesized using generative models (e.g., GANs or diffusion-based approaches) to approximate diverse typing-induced acoustic patterns without relying on real user data. Such synthetic augmentation may help preserve the benefits of dummy-user training while avoiding the collection or sharing of raw biometric signals, and could support more flexible and privacy-conscious deployment settings.

Environmental Sensitivity. Although ECHOType operates in the ultrasonic band and is largely isolated from audible noise, extreme environmental changes – such as highly reflective surfaces or strong airflow – may influence acoustic propagation. In our evaluation, stable performance was observed across typical office environments, suggesting that such effects are limited in everyday settings. Nevertheless, adaptive

thresholding or lightweight environment-aware calibration may further improve robustness in more diverse deployment contexts.

Advanced Replay and Signal Synthesis Attacks. While our evaluation demonstrates that simple replay attacks using prerecorded signals are ineffective, more sophisticated adversarial strategies can be considered. In particular, an attacker could attempt to synthesize or adaptively modify ultrasonic signals in real time to approximate the expected reflection patterns. Such attacks would require estimating the instantaneous phase and amplitude variations induced by finger motion and reproducing them accordingly. This entails modeling time-varying reflection paths and multi-path propagation characteristics, as well as maintaining precise synchronization with the user's typing behavior. Compared to simple replay, this represents a significantly more complex attack scenario. A systematic investigation of real-time signal synthesis and adaptive signal manipulation remains an important direction for future work.

10. Conclusion

In this paper, we presented ECHOType, a user identity recognition approach that utilizes ultrasonic reflections generated during everyday typing. ECHOType analyzes reflection patterns arising from the interaction between inaudible ultrasonic signals and users' finger motion, enabling identity recognition without altering natural typing behavior. Our experimental results show that ECHOType achieves an equal error rate (EER) of 0.031 in the single-user scenario, while in the multi-user scenario it achieves an F1 score of 0.940 and an unknown-user rejection accuracy of 95.5%. In addition, ECHOType remains effective under common variations such as keyboard types, accessory usage, and temporal changes, and shows resilience to multiple attack scenarios. Moreover, ECHOType can be implemented using only the built-in speaker and microphone of a commodity smartphone, without requiring specialized hardware or modifications to existing input devices. These results suggest that ECHOType provides a practical identity recognition solution and that typing-induced ultrasonic reflections can serve as an alternative biometric signal source for user identity recognition.

CRedit authorship contribution statement

Yena So: Writing – original draft, Software. **Youngjoo Park:** Writing – original draft, Software. **Chaehyeon Kim:** Software, Data curation. **Jihee Lee:** Software, Data curation. **Juhyeon Kim:** Software, Data curation. **Sunwoo Lee:** Writing – review & editing, Writing – original draft, Validation, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Ewha Womans University, South Korea Research Grant of 2026.

Appendix

Algorithm 1 ECHO_{TYPE} (Single-user Verification)

Require: Training signals D_{target} , auxiliary signals D_{dummy} , test signal $r(t)$

Ensure: Accept / Reject decision

```

1: // Enrollment and Training
2: for each signal in  $D_{\text{target}} \cup D_{\text{dummy}}$  do
3:   Capture reflected signal  $r(t)$ 
4:   Apply high-pass filter and compute spectrogram I
5: end for
6: Train CNN encoder  $\phi_\theta$  using multi-class classification
7: Extract feature representations p via softmax layer
8: Train binary SVM using p (target vs. dummy)
9: // Inference
10: Capture reflected signal  $r(t)$  from input
11: Preprocess to obtain spectrogram I
12:  $\mathbf{z} \leftarrow \phi_\theta(\mathbf{I})$ 
13:  $\mathbf{p} \leftarrow \text{softmax}(W_c \mathbf{z} + b_c)$ 
14:  $y \leftarrow \text{sign}(\mathbf{w}^T \mathbf{p} + b)$ 
15: return  $y$ 

```

Algorithm 2 ECHO_{TYPE} (Multi-user Identification)

Require: Training signals $\{D_i\}_{i=1}^N$, test signal $r(t)$

Ensure: User ID or *unknown*

```

1: // Enrollment and Training
2: for each user  $i$  do
3:   for each signal in  $D_i$  do
4:     Capture reflected signal  $r(t)$ 
5:     Apply high-pass filter and compute spectrogram I
6:   end for
7: end for
8: Train shared CNN encoder  $\phi_\theta$  using multi-class classification
9: Extract feature representations p via softmax layer
10: for each user  $i$  do
11:   Train binary SVM  $B_i$  for user  $i$ 
12: end for
13: // Inference
14: Capture reflected signal  $r(t)$  from input
15: Preprocess to obtain spectrogram I
16:  $\mathbf{z} \leftarrow \phi_\theta(\mathbf{I})$ 
17:  $\mathbf{p} \leftarrow \text{softmax}(W_c \mathbf{z} + b_c)$ 
18: for each user  $i$  do
19:    $g_i \leftarrow B_i(\mathbf{p})$ 
20: end for
21: if  $\max_i g_i > \tau_i$  then
22:    $\hat{i} \leftarrow \arg \max_i g_i$ 
23:   return  $\hat{i}$ 
24: else
25:   return unknown
26: end if

```

Data availability

The data that has been used is confidential.

References

- Acien, Alejandro, Morales, Aythami, Monaco, John V., Vera-Rodriguez, Ruben, Fierrez, Julian, 2021. TypeNet: Deep learning keystroke biometrics. *IEEE Trans. Biom. Behav. Identity Sci.* 4 (1), 57–70.
- Anand, S. Abhishek, Saxena, Nitesh, 2018. Keyboard emanations in remote voice calls: Password leakage and noise (less) masking defenses. In: *Proceedings of the Eighth ACM Conference on Data and Application Security and Privacy*. pp. 103–110.
- Bakelman, Ned, Monaco, John V., Cha, Sung-Hyuk, Tappert, Charles C., 2013. Keystroke biometric studies on password and numeric keypad input. In: *2013 European Intelligence and Security Informatics Conference*. pp. 204–207.
- Bendale, Abhijit, Boulton, Terrance E., 2016. Towards open set deep networks. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. CVPR*, pp. 1563–1572.
- Budzys, Arnoldas, Kurasova, Olga, Medvedev, Viktor, 2025. Integrating deep learning and data fusion for advanced keystroke dynamics authentication. *Comput. Stand. Interfaces* 92, 103931.
- Compagno, Alberto, Conti, Mauro, Lain, Daniele, Tsudik, Gene, 2017. Don't skype & type! acoustic eavesdropping in voice-over-IP. In: *Proceedings of the 2017 ACM Asia Conference on Computer and Communications Security. AsiaCCS*, pp. 703–715.
- Giallanza, Tyler, Siems, Travis, Smith, Elena, Gabrielsen, Erik, Johnson, Ian, Thornton, Mitchell A., Larson, Eric C., 2019. Keyboard snooping from mobile phone arrays with mixed convolutional and recurrent neural networks. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 3 (2), 1–22.
- Giot, Romain, El-Abed, Mohamad, Rosenberger, Christophe, 2009. GREYC keystroke: A benchmark for keystroke dynamics biometric systems. In: *2009 IEEE 3rd International Conference on Biometrics: Theory, Applications, and Systems*. pp. 1–6.
- Giot, Romain, El-Abed, Mohamad, Rosenberger, Christophe, 2012. Web-based benchmark for keystroke dynamics biometric systems: A statistical analysis. In: *2012 Eighth International Conference on Intelligent Information Hiding and Multimedia Signal Processing*. pp. 11–15.
- Grabham, N. J., White, N. M., 2008. Use of a novel keypad biometric for enhanced user identity verification. In: *2008 IEEE Instrumentation and Measurement Technology Conference*. pp. 12–16.
- Halevi, Tzipora, Saxena, Nitesh, 2012. A closer look at keyboard acoustic emanations: Random passwords, typing styles and decoding techniques. In: *Proceedings of the 7th ACM Symposium on Information, Computer and Communications Security*. pp. 89–90.
- Huang, Jinyang, Bai, Jia-Xuan, Zhang, Xiang, Liu, Zhi, Feng, Yuanhao, Liu, Jianchun, Sun, Xiao, Dong, Mianxiong, Li, Meng, 2024. KeystrokeSniffer: An off-the-shelf smartphone can eavesdrop on your privacy from anywhere. *IEEE Trans. Inf. Forensics Secur.*
- Killourhy, Kevin S., Maxion, Roy A., 2009. Comparing anomaly-detection algorithms for keystroke dynamics. In: *2009 IEEE/IFIP International Conference on Dependable Systems and Networks. DSN*, pp. 125–134.
- Kim, Yooshin, Kwon, Namhyeok, Shin, Donghoon, 2025. KDPrint: Passive authentication using keystroke dynamics-to-image encoding via standardization. *Comput. Secur.* 104725.
- Medvedev, Viktor, Budzys, Arnoldas, Kurasova, Olga, 2025. A decision-making framework for user authentication using keystroke dynamics. *Comput. Secur.* 104494.
- Muda, Lindasalwa, Begam, Mumtaj, Elamvazuthi, Irraivan, 2010. Voice recognition algorithms using mel frequency cepstral coefficient (MFCC) and dynamic time warping (DTW) techniques. *arXiv preprint arXiv:1003.4083*.
- Nyquist, Harry, 1928. Certain topics in telegraph transmission theory. *Trans. Am. Inst. Electr. Eng.* 47 (2), 617–644.
- Shadman, Rashik, Wahab, Ahmed Anu, Manno, Michael, Lukaszewski, Matthew, Hou, Daqing, Hussain, Faraz, 2025. Keystroke dynamics: Concepts, techniques, and applications. *ACM Comput. Surv.* 57 (11), 1–35.
- Statista Research Department, 2025. Distribution of internet users worldwide as of february 2024, by age group. <https://www.statista.com/statistics/272365/age-distribution-of-internet-users-worldwide/>. [Online; Accessed 18 June 2025].
- Stragapede, Giuseppe, Delgado-Santos, Paula, Tolosana, Ruben, Vera-Rodriguez, Ruben, Guest, Richard, Morales, Aythami, 2024. TypeFormer: Transformers for mobile keystroke biometrics. *Neural Comput. Appl.* 36 (29), 18531–18545.
- Wang, Jian, Ruby, Rukhsana, Wang, Lu, Wu, Kaishun, 2016. Accurate combined keystrokes detection using acoustic signals. In: *2016 12th International Conference on Mobile Ad-Hoc and Sensor Networks. MSN*, pp. 9–14.
- WordsRated, 2024. Typing speed statistics. <https://wordsrated.com/typing-speed-statistics/>. [Online; Accessed 26 June 2024].
- Zhao, Yanchao, Zhao, Yiming, Li, Si, Han, Hao, Xie, Lei, 2023. UltraSnoop: Placement-agnostic keystroke snooping via smartphone-based ultrasonic sonar. *ACM Trans. Internet Things* 4 (4), 1–24.



Yena So is an undergraduate student in the Department of Information Security at Seoul Women's University, Seoul, South Korea. Her research interests include AI for security, and Privacy Enhancing Technologies (PET).



YoungJoo Park is an undergraduate student in the Department of Information Security at Seoul Women's University, Seoul, South Korea. Her research interests include IoT security, AI for security, and digital forensics.



Chaehyeon Kim is an undergraduate student in the Department of Information Security at Seoul Women's University, Seoul, South Korea. Her research interests include blockchain, cryptography, and pen testing.



Jihee Lee is an undergraduate student in the Department of Information Security at Seoul Women's University, Seoul, South Korea. Her research interests include IoT security, and cryptology.



Juhyeon Kim is an undergraduate student in the Department of Information Security at Seoul Women's University, Seoul, South Korea. Her research interests include digital forensics, mobile forensics, and privacy protection.



Sunwoo Lee received the B.S. degree in mathematics from Sogang University, Seoul, South Korea, in 2015, and the Ph.D. degree in information security from Korea University, Seoul, South Korea, in 2022. After completing her Ph.D., she worked as a Research Professor at the Graduate School of Information Security, Korea University, until September 2022. From October 2022 to February 2024, she was a Staff Engineer at Samsung Research, Seoul, South Korea. She then joined the Department of Information Security at Seoul Women's University, Seoul, South Korea, as an Assistant Professor, where she served from February 2024 to February 2025. Since March 2026, she has been with Ewha Womans University, Seoul, South Korea, as an Assistant Professor. Her research interests include side-channel attacks, data-driven security, IoT security, sensor security, and usable security.